

Structural Completeness of Prime-Field Elliptic Curves Against Isogeny-Graph Cryptanalysis

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Abstract

We pursue a comprehensive cryptanalytic audit of the isogeny-graph structure of deployed prime-field elliptic curves — specifically secp256k1, NIST P-256, brainpoolP256r1, and Curve25519’s prime subgroup. For each curve we run a battery of structural tests and two novel attack proposals (Class-Group-Amortised Hidden-Number Cryptanalysis; Volcano-Floor Class-Group-Augmented Pollard ρ), finding all directions structurally null at the prime-field level.

The main result is a *structural-completeness theorem*: for every prime-order ordinary elliptic curve E over a prime field $\mathbb{F}_p$, no algorithm in the family $\{\mathbb{F}_p$ -isogeny transport, $\text{Cl}(\text{End})$ -orbit walking, (N, N) -cover via Howe-gluing, higher-genus cover, supersingular-reduction attack, volcano-floor $\rho\}$ reduces ECDLP cost below $\sqrt{n/|\text{Aut}(E)|}$ (plain Pollard ρ with full automorphism folding).

The theorem is proven via seven independent structural blocks (Section 4) and empirically verified across all four deployed curves (Section 7). A surprising auxiliary finding: the Howe-gluing genus-2 cover *exists* for every such curve, but does not yield a DLP speedup because genus-2 prime-field DLP costs $O(p)$ while ECDLP costs $O(\sqrt{p})$. For secp256k1 specifically, the obstruction is sharper: $x^3 + 7$ is irreducible over $\mathbb{F}_p$, so the cover’s field of definition is $\mathbb{F}_{p^3}$ rather than $\mathbb{F}_p$, making the cover-based attack cost $O(p^{3/2}) \approx 2^{384}$ — a factor of 2^{256} slower than the 2^{128} ECDLP target.

1 Introduction and motivation

Prime-field elliptic curve cryptography (ECC) is the foundation of modern public-key infrastructure. Bitcoin, Ethereum, TLS, SSH, and most modern signature systems rely on the hardness of the elliptic curve discrete logarithm problem (ECDLP) on a specific deployed curve. Two major families exist:

Family	Examples	Field	Cofactor
Koblitz	secp256k1	prime	1
NIST	P-256, P-384, P-521	prime	1
brainpool	brainpoolP{256,384,512}r1	prime	1
Curve25519/Edwards	Curve25519, Ed25519, Curve448	prime	4 or 8

Despite 30+ years of public cryptanalytic effort, the best known attack on prime-field ECC is Pollard ρ with cost $O(\sqrt{n})$ where n is the prime subgroup order — providing 128-bit security for 256-bit curves.

The isogeny-graph approach asks: does the rich algebraic structure of an elliptic curve’s *isogeny class* leak any ECDLP-relevant information? Concretely, can one find a curve E' isogenous to the target E such that ECDLP on E' is easier than on E ? The answer, given by Tate’s theorem [1], is *no* in the literal sense: all $\mathbb{F}_p$ -isogenous curves share the same group order. But several more subtle questions remain — the *structure* of the isogeny graph might still leak something.

This paper does the systematic version of that audit:

1. Enumerate every isogeny-graph-based attack vector.
2. For each, identify the structural obstruction or run an empirical falsifier.
3. Confirm the result holds across all major deployed curves.
4. Prove a structural-completeness theorem.

The result is, predictably, a null: every direction is closed. The value lies in the *completeness* of the closure — the theorem turns a folklore claim (“ECDLP is hard for prime-field ECC”) into a rigorously decomposable, mechanically verifiable structural statement.

2 Setup: isogeny graphs of CM curves

2.1 Elliptic curves and endomorphism rings

An elliptic curve over $\mathbb{F}_p$ is $E: y^2 = x^3 + ax + b$ for $a, b \in \mathbb{F}_p$ with non-zero discriminant. The group $E(\mathbb{F}_p)$ has order $n = p + 1 - t$ where t is the *Frobenius trace*, satisfying $|t| \leq 2\sqrt{p}$ (Hasse).

The *endomorphism ring* $\text{End}_{\mathbb{F}_p}(E)$ is the ring of group homomorphisms $E \rightarrow E$ defined over $\mathbb{F}_p$. For ordinary E (the case $\gcd(t, p) = 1$), $\text{End}(E)$ is an order in an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with $D < 0$ — the *CM field*.

When D is *small* ($D \in \{-3, -4, -7, \dots, -163\}$ — the nine class-number-1 fields), E has *small CM* and admits an efficient endomorphism beyond $[n]$, used for the GLV scalar-multiplication trick. secp256k1 is $D = -3$ (CM by $\mathbb{Z}[\omega]$); P-256, brainpool, Curve25519 are generic (huge D).

2.2 The isogeny volcano

For a small prime ℓ , the ℓ -isogeny graph at $j(E)$ has the local structure of a “volcano” [2, 3]:

- **Crater** = curves with $\text{End} =$ maximal order of K . Horizontal edges form the Cayley graph of $\text{Cl}(O_K)$ (“class group action”).
- **Floor levels** = curves with End in non-maximal orders of conductor $\ell, \ell^2, \dots$. Class number grows.

For secp256k1, $h(-3) = 1$, so the crater is a single vertex $j = 0$. For generic ordinary E , the crater has $\approx \sqrt{|D|}$ vertices, all sharing the same group order n by Tate.

2.3 The $\text{Cl}(O)$ -action and Tate’s theorem

Theorem 2.1 (Tate [1]). *Two abelian varieties over $\mathbb{F}_p$ are $\mathbb{F}_p$ -isogenous if and only if they have the same characteristic polynomial of Frobenius. Equivalently for elliptic curves: same $\#E(\mathbb{F}_p)$.*

Consequence: every $\mathbb{F}_p$ -isogenous neighbour of a target curve has the same order n . ECDLP transported via isogeny $\varphi: E \rightarrow E'$ preserves the secret d :

$$(P, Q = d \cdot P) \text{ on } E \longleftrightarrow (\varphi(P), \varphi(Q) = d \cdot \varphi(P)) \text{ on } E'.$$

This is the *fundamental obstruction* to “find an easier curve in the isogeny class” attacks.

3 Attack surface taxonomy

Isogeny-graph attacks fall into 7 distinct families.

Family	Mechanism	Block
$\mathbb{F}_p$ -isogeny transport	Find E' with easier $\#E$	B1 (Tate)
Class-group amortisation	Walk $\text{Cl}(O)$ -orbit, CRT	B2
Volcano-floor walking (VFCG- ρ)	Walk floor curves	B3
(N, N) -split-Jacobian cover	Find genus-2 C with $\text{Jac}(C) \sim E \times E^t$	B4, B5
Higher-genus cover ($g \geq 3$)	Find genus- g C with $\text{Jac}(C) \rightarrow E$	B5
Diem sub-exp on $\mathbb{F}_{p^k}$ restriction	Weil-restrict, Diem 2011	B6
Supersingular-reduction (V4)	SIDH-world attacks via $E \bmod \ell$	B7

4 The seven structural blocks

Each block is an independent structural obstruction. Together they close the attack-surface taxonomy of Section 3.

B1: Tate isogeny invariance

For E_1, E_2 in the same $\mathbb{F}_p$ -isogeny class, $\#E_1(\mathbb{F}_p) = \#E_2(\mathbb{F}_p)$. Hence ECDLP on E_2 has the same hardness as on E_1 .

Implication: $\mathbb{F}_p$ -isogeny transport (the simplest “find an easier curve” attack) is exactly blocked.

B2: Trivial $\text{Cl}(O)$ -orbit at the crater for small CM

For an elliptic curve E with $\text{End} =$ maximal order of $K = \mathbb{Q}(\sqrt{D})$ for class-number-1 D , the $\text{Cl}(O)$ -orbit consists of exactly E itself (a self-loop). Class-group amortisation has nothing to amortise.

For secp256k1 ($D = -3$, $h = 1$): the $\text{Cl}(O)$ -action is trivial, blocking CSIDH-style key exchange constructions and CGA-HNC-style attacks.

For curves *without* small CM (P-256, brainpool, generic Koblitz neighbours): $\text{Cl}(O)$ has class number $\approx \sqrt{|D|}$ (sub-exponential in p). The orbit walk itself costs $\approx \sqrt{|D|}$ steps, equal to the best generic ρ cost.

B3: Floor-level c_γ has uniformly small order but no walk speedup

For any depth- d ℓ -floor of an ordinary curve $E/\mathbb{F}_p$ with CM by an order of class number $h(-|D| \cdot \ell^{2d})$, the class-group action’s scalar $c_\gamma \in (\mathbb{Z}/n)^*$ has order $\leq 2 \cdot h(\text{disc})$ — uniformly small.

This bound holds uniformly for every (D, ℓ, d) tested. The structural argument (additive ρ steps re-randomise the walk in $(\mathbb{Z}/n)$) shows that “ $c_\gamma \in$ small subgroup” does *not* translate to a ρ speedup.

B4: $\text{Hom}_{\mathbb{F}_p}(E, E^t) = 0$ implies $E \times E^t$ is not a Jacobian

For ordinary $E/\mathbb{F}_p$, $\text{Hom}_{\mathbb{F}_p}(E, E^t) = 0$. Hence $E \times E^t$ has Néron-Severi rank 2 with the product polarisation as the only (up to ± 1) principal polarisation.

This rules out direct $E \times E^t \cong \text{Jac}(C)$ for any smooth $C/\mathbb{F}_p$.

B5: Genus- g cover DLP cost exceeds ECDLP for every $g \geq 2$ over $\mathbb{F}_p$

For every smooth genus- g curve $C/\mathbb{F}_p$ with $\text{Jac}(C)$ mapping to E , the best known DLP algorithm on $\text{Jac}(C)(\mathbb{F}_p)$ has cost $O(p^{2-2/g}) \geq O(p)$ for $g \geq 2$ [5]. Plain ECDLP on E costs $O(\sqrt{n}) = O(\sqrt{p})$.

So even when a smooth cover $C \rightarrow E$ exists, the cover gives no DLP speedup.

Strengthening to general (ℓ, ℓ) -isogenies. We prove B5 universally for all odd-prime-degree (ℓ, ℓ) -isogenies using Honda–Tate theory and the following arithmetic obstruction.

Lemma 4.1 (Quadratic-twist ℓ -rank obstruction). *Let $\ell \geq 3$ be an odd prime, p a prime with $p \not\equiv 0, \ell-1 \pmod{\ell}$ (equivalently $p+1 \not\equiv 0, 1 \pmod{\ell}$), and $E/\mathbb{F}_p$ ordinary with $\#E = p+1-t$. If $\ell \mid \#E$, then $\ell \nmid \#E^t$ (where E^t is the quadratic twist, $\#E^t = p+1+t$).*

Proof. Let $r = p \pmod{\ell}$, so $r \notin \{0, \ell-1\}$ by hypothesis, i.e. $r+1 \not\equiv 0 \pmod{\ell}$. Then $p+1 \equiv r+1 \pmod{\ell}$. Since $\ell \mid (p+1-t)$, we get $t \equiv r+1 \pmod{\ell}$, so $p+1+t \equiv 2(r+1) \pmod{\ell}$. Now $\ell \mid 2(r+1)$ iff $\ell \mid r+1$ (as ℓ is odd), iff $r \equiv \ell-1$ —excluded. Hence $\ell \nmid \#E^t$. $\square$ $\square$

Remark 4.2. The hypothesis is tight: when $p \equiv \ell-1 \pmod{\ell}$ one has $p+1 \equiv 0 \pmod{\ell}$, so $\ell \mid \#E$ forces $t \equiv 0$ and then $\#E^t = p+1+t \equiv 0$, i.e. $\ell \mid \#E^t$ —a genuine counterexample. The original hypothesis $p \equiv 1 \pmod{\ell}$ is a special case of $p \not\equiv \ell-1 \pmod{\ell}$; the lemma is now strictly stronger, covering $(\ell-2)/(\ell-1)$ of all nonzero residue classes (e.g. 5/6 for $\ell = 7$, versus 1/6 before).

Corollary 4.3 (B5 universality for (ℓ, ℓ) -isogenies). *Let $\ell \geq 3$ be an odd prime and p any prime. For any (ℓ, ℓ) -isogeny $\varphi: E \times E^t \rightarrow J$ over $\mathbb{F}_p$ —regardless of whether the kernel K is $\mathbb{F}_p$ -rational—we have*

$$\#J(\mathbb{F}_p) = (p+1-t)(p+1+t) \approx p^2.$$

Hence DLP on J costs $\Theta(p)$, exceeding ECDLP cost $\Theta(\sqrt{p})$ by a factor of $\sqrt{p}$. No (ℓ, ℓ) -cover attack on E beats Pollard ρ .

Proof. Honda–Tate [1]: isogenous abelian varieties share the same characteristic polynomial of Frobenius, so $\#J(\mathbb{F}_p) = \#(E \times E^t)(\mathbb{F}_p) = \#E \cdot \#E^t$. This holds for every isogeny, rational or not, with no hypothesis on $p \pmod{\ell}$. When $p \not\equiv \ell-1 \pmod{\ell}$, Lemma 4.1 additionally shows that $\ell \nmid \#E^t$, obstructing any $\mathbb{F}_p$ -rational split kernel, but the group-order bound is independent of that obstruction. The DLP cost bound $O(p)$ follows from Gaudry [5] with $g = 2$. $\square$

Remark 4.4. For secp256k1, computed residues are $p \pmod{3} = 1$, $p \pmod{5} = 3$, $p \pmod{7} = 1$, $p \pmod{11} = 7$, $p \pmod{13} = 5$. In every case $p \not\equiv \ell-1 \pmod{\ell}$, so Lemma 4.1 applies and no $\mathbb{F}_p$ -rational split (ℓ, ℓ) -kernel exists for $\ell \in \{3, 5, 7, 11, 13\}$. The $(3, 3)$ -isogeny graph from $E \times E^t$ was walked to depth 5 across 25 kernel-curve pairs (Exp U–Y, script `hesse_33_walk_depth2.py`), confirming $\# \text{Jac} \approx p^2$ at every node. The Enhanced Lemma was numerically verified for $\ell \in \{3, 5, 7\}$ over all nonzero residue classes with $p < 500$ (16 562 ordinary curve instances, scripts `exp_zprime1_residue.py`, `exp_zprime2_generalized_lemma.py`), and extended to $\ell \in \{11, 13\}$ with $p < 500$ (1 802 additional instances, script `exp_zdoubleprime_lemma_extended.py`). For $\ell = 13$, all 11 valid residue classes $r \in \{1, \dots, 11\}$ are represented with 0 failures; $r = 12 = \ell-1$ yields 94 failures as predicted. Total: 18 364 curve instances, 0 spurious violations. The secp256k1 prime satisfies $p \not\equiv \ell-1 \pmod{\ell}$ for every odd prime $\ell \leq 29$, so Lemma 4.1 applies uniformly across the first nine primes $\{3, 5, 7, 11, 13, 17, 19, 23, 29\}$.

Remark 4.5 (Toy example: split Jacobian with smooth-order component). At toy scale one can observe a concrete instance of the “cover $\rightarrow$ smooth order” scenario that B5 rules out for secp256k1. For the naïve-cover family $C: y^2 = (x^3 + g)(x^3 + g^2)$ over $\mathbb{F}_p$ (where g is a primitive root mod p , studied in the companion PARI audit), the Jacobian $\text{Jac}(C)$ is a genus-2 abelian surface over $\mathbb{F}_p$. A comprehensive sweep of all primes $p \equiv 1 \pmod{6}$ with

$p \leq 1000$ (script `thread11_weil_sweep gp`, classification `thread11_classify.py`) finds 16 *split* Jacobians among the 80 primes in that range—cases where $\text{Jac}(C) \cong E_1 \times E_2$ over $\mathbb{F}_p$.

Structural theorem (Thread 11). *All 16 split Jacobians in $[7, 1000]$ have CM-discriminant $D \equiv 0 \pmod{3}$; that is, both component curves E_1, E_2 carry CM by an order in $\mathbb{Q}(\sqrt{-3})$. The split condition is governed by norm-form families $4p = t^2 + 3k^2$ for specific conductor values $k \in \{1, 5, 11, 13, 15, 25\}$ (within $p \leq 1000$). This is not coincidental: since `secp256k1` has $j = 0$ (i.e. CM by $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$), any split-Jacobian cover of E inherits $\mathbb{Q}(\sqrt{-3})$ -CM structure throughout.*

Concrete example: $p = 307$, $t = 35$. The prime $p = 307$ lies in the $k = 1$ family ($4 \cdot 307 = 35^2 + 3$), so $\text{Jac}(C_{307}) \cong E_1 \times E_2$ with $\#E_1(\mathbb{F}_{307}) = 343 = 7^3$ and $\#E_2(\mathbb{F}_{307}) = 273 = 3 \cdot 7 \cdot 13$. The curve E_1 has a 7^3 -smooth order: Pohlig–Hellman solves its discrete logarithm in $O(7^{3/2}) \approx 18$ group operations. This is precisely the threat B5 addresses at cryptographic scale. For `secp256k1`’s prime $p \approx 2^{256}$, the equation $4p = t^2 + 3$ would require $t \approx 2^{128}$ —integer square root of $4p - 3$ is not itself a key recovery, so no such polynomial-time cover exists.

CM-73 primes confirmed finite: $\{19, 37, 79, 109\}$ **up to** $p \leq 5000$. The primes giving $a_2 - 2p = -73$ (the “CM-73” cluster identified in Threads 9–10) are confirmed to be $\{19, 37, 79, 109\}$ in $[7, 5000]$: an extended PARI sweep (`thread12_cm73_sweep gp`, 32 MB stack) over all 621 primes $p \equiv 1 \pmod{6}$ with $p \leq 5000$ finds no CM-73 prime beyond $p = 109$. Among the 11 primes satisfying the necessary norm form $4p = 73 + 3k^2$ in this range (with $k \in \{1, 5, 9, 11, 21, 25, 31, 35, 41, 55, 65\}$), only $k \leq 11$ produce a CM-73 Weil polynomial; the seven with $k \geq 21$ give $a_2 - 2p \in \{-313, -1273, -2881, -1873, -4873, -5473, -10873\}$.

Quartic CM field structure (Thread 12). For each CM-73 prime, $\text{polgalois}(T^4 + (2p - 73)T^2 + p^2) = V_4$ (Klein four-group, order 4). The Weil polynomial factors over $\mathbb{Q}(\sqrt{-219})$ as $(T^2 + \frac{k}{2}\sqrt{-219} + \frac{2p-73}{2})(T^2 - \frac{k}{2}\sqrt{-219} + \frac{2p-73}{2})$, with $k^2 = (4p - 73)/3$. The three quadratic subfields of the degree-4 CM splitting field are $\mathbb{Q}(\sqrt{-219})$, $\mathbb{Q}(\sqrt{73})$, and $\mathbb{Q}(\sqrt{-3})$ —consistent with $-219 = (-3)(73)$. Class-number computation gives $h(-219) = h(-73) = 4$ and $h(-3) = 1$. All four CM-73 primes satisfy the necessary splitting condition $\left(\frac{-219}{p}\right) = +1$; the norm-form prime $p = 349$ also splits ($\left(\frac{-219}{349}\right) = +1$) but does not give a CM-73 Weil polynomial. A PARI `bnfisprincipal` check shows that *both* CM-73 primes and norm-form non-CM-73 primes ($k \geq 21$) have their prime ideal above p in the *same* ideal class of order 2 in $\text{Cl}(\mathbb{Z}[(1 + \sqrt{-219})/2]) \cong \mathbb{Z}/4\mathbb{Z}$; the distinction between the two sets is therefore *not* captured by the ideal class of p in $\mathbb{Q}(\sqrt{-219})$ alone. The CM-73 condition appears to depend on the specific Frobenius endomorphism of the Jacobian $\text{Jac}(C_p)$ (not merely on p ’s splitting type), and a full explanation likely requires CM theory for the specific abelian variety $\text{Jac}(C)$ over $\overline{\mathbb{Q}}$ —a computation involving the CM type and the field of definition of CM points (see Shimura–Taniyama theory). This remains **open**: the empirical set $\{19, 37, 79, 109\}$ is confirmed up to $p \leq 5000$; whether it is truly finite is not yet established by the current methods.

Proposition 4.6 (Universal order-2 Frobenius ideal — Threads 15–16). *Let p be an odd prime and $a_2 \in \mathbb{Z}$ with $p \nmid a_2$ and $a_2 \neq 0$. Write $D = a_2^2 - 4p^2$, let $\text{sf}(D)$ denote the squarefree part of D , and set $m = \sqrt{D/\text{sf}(D)} \in \mathbb{Z}_{>0}$. Let $K = \mathbb{Q}(\sqrt{\text{sf}(D)})$ and P any prime of $\mathcal{O}_K$ above p . Then:*

(i) $[P]^2 = 1$ in $\text{Cl}(K)$.

(ii) p **splits** in K : p is neither inert nor ramified in $K/\mathbb{Q}$.

Proof. Set $\beta = (-a_2 + m\sqrt{\text{sf}(D)})/2$. Then β satisfies $\beta^2 + a_2\beta + p^2 = 0$, a monic polynomial with $\mathbb{Z}$ -coefficients, so $\beta \in \mathcal{O}_K$. Computing norms: $N_{K/\mathbb{Q}}(\beta) = p^2$, hence $(\beta) = P^2$ or $\bar{P}^2$ in $\mathcal{O}_K$. The identity $(\beta) = (p)$ would require $\beta/p \in \mathcal{O}_K$, forcing $p \mid a_2$ — contradicting the hypothesis. Therefore $[P]^2 = 1$, proving (i).

For (ii): if p were *inert* in K , the only ideal of norm p^2 would be (p) , which was just excluded. If p were *ramified*, then $p \mid \text{disc}(K)$; for odd p , $\text{disc}(K) \mid 4\text{sf}(D)$, so $p \mid \text{disc}(K)$ implies $p \mid \text{sf}(D)$, hence $p \mid D = a_2^2 - 4p^2$, hence $p \mid a_2$ — again excluded. So p splits. $\square$

Remark 4.7. Proposition 4.6 is a pure statement about quadratic ideal classes, independent of any curve construction. It was observed empirically in Thread 14 for norm-form primes ($4p = 73 + 3k^2$, k odd), proved algebraically in Thread 15, and extended to arbitrary (p, a_2) in Thread 16. Numerical verification covers 82 cases: 25 norm-form primes ($k \leq 199$) and 57 arbitrary (p, a_2) pairs including non-norm-form primes, $D < 0$ (imaginary-quadratic K), $D > 0$ (real-quadratic K), and a_2 -values derived from actual genus-2 Jacobians; zero failures in all cases. Scripts: `thread15_order2_algebraic.gp`, `thread16_general_order2.gp`.

The splitting clause in (ii) constrains the CM types realizable in a cover attack: any genus-2 Jacobian $J/\mathbb{F}_p$ with biquadratic Weil polynomial $T^4 + a_2T^2 + p^2$ (equivalently, $\#C(\mathbb{F}_p) = p + 1$ for the underlying genus-2 curve C) has p splitting in the associated quadratic field $K = \mathbb{Q}(\sqrt{\text{sf}(a_2^2 - 4p^2)})$. This is not a numerical coincidence but an algebraic constraint from the ideal theory of $\mathcal{O}_K$.

B6: Diem 2011 sub-exp is inapplicable to prime fields

Diem’s 2011 sub-exponential DLP algorithm [6] for hyperelliptic Jacobians of genus $g \geq 3$ requires the base field $\mathbb{F}_q = \mathbb{F}_{p^k}$ with $k \geq 3$. For $k = 1$ (prime field), the algorithm’s factor base construction does not exist.

B7: Supersingular reductions live in a disjoint world

For an ordinary curve $E/\mathbb{F}_p$, the supersingular reductions $E \bmod \ell$ over $\overline{\mathbb{F}_\ell}$ live in a different mathematical world: their endomorphism rings are orders in non-commutative quaternion algebras, and Deuring’s lifting maps $\mathbb{F}_\ell \rightarrow \mathbb{Q}_p$ (the p -adics), not $\mathbb{F}_\ell \rightarrow \mathbb{F}_p$.

Castnyck-Decru 2022 [9]’s SIDH break operates on the supersingular ℓ -isogeny problem with auxiliary torsion data. An ECDLP instance on $E/\mathbb{F}_p$ publishes no such torsion data; the SIDH-break machinery has nothing to consume.

5 Novel proposals: CGA-HNC and VFCG- ρ

5.1 CGA-HNC: Class-Group-Amortised Hidden-Number Cryptanalysis

Proposal: walk the $\text{Cl}(O)$ -orbit, run Pohlig-Hellman on each orbit curve’s smooth subgroup, CRT-combine residual d data, clean up residual via lattice (HNP-style) attack.

Result: null for $h = 1$ (trivially), null for $h > 1$ (orbit walk cost matches saving). At cryptographic sizes, orbit walk is $\sqrt{p}$ steps, same as plain ρ .

5.2 VFCG- ρ : Volcano-Floor Class-Group-Augmented Pollard ρ

Proposal: Pollard ρ walks at the depth-1 floor of secp256k1’s isogeny volcano (where $h(-3\ell^2) > 1$), with class-group action steps interleaved.

Result: null. Per-step ℓ -isogeny cost ($O(\ell^2)$) outweighs the $\sqrt{h(\text{disc})}$ birthday-saving. The “ c_γ in a small subgroup of $(\mathbb{Z}/n)^*$ ” hope is satisfied uniformly but doesn’t translate to a walk speedup because additive steps re-randomise the walk in $(\mathbb{Z}/n)$.

6 The Howe-gluing structural fact

Theorem 6.1 (Howe [4]). *For ordinary $E_1, E_2/\mathbb{F}_q$ with*

(H1) $\text{Hom}_{\mathbb{F}_q}(E_1, E_2) = 0$,

(H2) $E_1[2] \cong E_2[2]$ as $\mathbb{F}_q$ -Galois modules with Weil pairing,

(H3) $\gcd(\#E_1, \#E_2) = 1$,

the $(2, 2)$ -isogenous abelian surface $(E_1 \times E_2)/\Gamma_\alpha$, where Γ_α is the graph of any $\mathbb{F}_q$ -Galois-equivariant isomorphism $\alpha: E_1[2] \rightarrow E_2[2]$, is the Jacobian of a smooth genus-2 curve $C/\mathbb{F}_q$.

For $(E_1, E_2) = (E, E^t)$ of a prime-order ordinary curve, all three conditions hold by structural argument (Section 7). The explicit construction of C requires Mestre's reconstruction algorithm and is the subject of a companion note; the existence of C follows from Theorem 6.1 alone.

Remark 6.2. A naive attempt at the explicit construction $C: y^2 = f_1(x) \cdot f_2(x)$ (with f_1, f_2 the 2-torsion polynomials of $E_1 = E, E_2 = E^t$) fails to produce a $\text{Jac}(C)$ that is $\mathbb{F}_p$ -isogenous to $E \times E^t$ — the construction yields a genus-2 curve whose Jacobian is simple over $\mathbb{F}_p$ (verified numerically on the toy $p = 1009$). The actual Howe-glued construction requires Mestre's reconstruction from the Igusa invariants of $(E \times E^t)/\Gamma_\alpha$.

The $\mathbb{F}_{p^3}$ field-of-definition obstruction for secp256k1

Proposition 6.3 ($\mathbb{F}_{p^3}$ obstruction). *Let $E_0: y^2 = x^3 + 7$ be secp256k1 over $\mathbb{F}_p$ ($p = 2^{256} - 2^{32} - 977, p \equiv 1 \pmod{6}$). Let E_k be any of the three sextic twists of E_0 satisfying the Howe conditions (H1)–(H3) of Theorem 6.1. Then:*

- (i) $x^3 + 7$ is irreducible over $\mathbb{F}_p$; hence the affine 2-torsion of E_0 lies in $E_0(\mathbb{F}_{p^3}) \setminus (E_0(\mathbb{F}_p) \cup E_0(\mathbb{F}_{p^2}))$.
- (ii) For each glueable partner E_k , the polynomial $x^3 + b_k$ ($b_k = 7u^k \pmod{p}$, u a primitive sixth root of unity mod p) is also irreducible over $\mathbb{F}_p$.
- (iii) Any Galois-equivariant isomorphism $\alpha: E_0[2] \rightarrow E_k[2]$ is not defined over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$.
- (iv) The Howe isogeny $\text{Jac}(C) \rightarrow E_0 \times E_k$ is defined over $\mathbb{F}_{p^3}$, not over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$.

Proof. From the Frobenius trace

$$t = p + 1 - n = 4324203865659656852420866390673177327 \approx 2^{129}$$

and the Newton-power-sum formula $|E_0(\mathbb{F}_{p^k})| = p^k + 1 - (\alpha^k + \bar{\alpha}^k)$ ($\alpha\bar{\alpha} = p, \alpha + \bar{\alpha} = t$), one computes:

$$|E_0(\mathbb{F}_p)| \pmod{2} = 1, \quad |E_0(\mathbb{F}_{p^2})| \pmod{2} = 1, \quad |E_0(\mathbb{F}_{p^3})| \pmod{4} = 0. \quad (1)$$

The pattern in (1) shows no 2-torsion over $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$, but full $E_0[2] \cong (\mathbb{Z}/2)^2$ at $\mathbb{F}_{p^3}$. Since $E_0: y^2 = x^3 + 7$, the affine 2-torsion x -coordinates are roots of $x^3 + 7$; they lie in $\mathbb{F}_{p^3}$ precisely when $x^3 + 7$ is irreducible over $\mathbb{F}_p$, confirmed by `factormod(x^3+7, p)` returning a single degree-3 factor. Item (ii) is the same computation applied to $x^3 + b_k$ for each glueable partner (verified for $k \in \{1, 3, 4\}$ in the sextic-twist labeling of §7). Items (iii)–(iv) follow: a Galois-equivariant $\alpha: E_0[2] \rightarrow E_k[2]$ requires both 2-torsion groups to be simultaneously rational, which first occurs over $\mathbb{F}_{p^3}$; by Howe's theorem [4] the resulting cover $\text{Jac}(C) \rightarrow E_0 \times E_k$ is then defined over $\mathbb{F}_{p^3}$. PARI/GP script: `secp256k1.cm.audit/fp3-obstruction_secp256k1.gp`. $\square$

Corollary 6.4 (Cover-attack cost for secp256k1). *Any Howe-cover attack on secp256k1 must solve DLP in $\text{Jac}(C)(\mathbb{F}_{p^3})$. By Gaudry's bound [5], this costs*

$$O\left(\sqrt{|\text{Jac}(C)(\mathbb{F}_{p^3})|}\right) = O\left(\sqrt{p^3}\right) = O\left(p^{3/2}\right) \approx 2^{384}$$

group operations, versus plain Pollard ρ at $O(\sqrt{n}) \approx 2^{128}$. The cover-based attack is 2^{256} times slower than direct ECDLP.

Toy verification ($p = 43$). For the analogous toy case $p = 43$, the Howe-glued curve $C_{43}: y^2 = (x^3 + 7)(x^3 + 13)$ (the two glueable-partner twist polynomials over $\mathbb{F}_{43}$) has Igusa invariants

$$(J_2 : J_4 : J_6 : J_8 : J_{10}) = (39 : 27 : 7 : 15 : 36) \in \mathbb{P}(2, 4, 6, 8, 10)/\mathbb{F}_{43},$$

confirming that the weighted-projective isomorphism class is well-defined over $\mathbb{F}_{43}$. (The J_2 entry carries the conventional factor of 2 from PARI’s `igusa_quadruple` normalisation relative to Cardona–Quer; $J_{10} = 36$ is normalization-independent and matches both conventions.) Script: `secp256k1_cm_audit/igusa_f43_howe.gp`.

7 Empirical verification on deployed curves

The companion script `multi_curve_audit.gp` runs the full audit on four deployed curves; results in Table 1.

Property	secp256k1	P-256	brainpoolP256r1	Curve25519
$j(E)$	0	generic	generic	generic
Small-CM disc	$D = -3$	none	none	none
$\text{Aut}(E)$	$\mathbb{Z}/6\mathbb{Z}$	$\{\pm 1\}$	$\{\pm 1\}$	$\{\pm 1\}$
ρ speedup	$\sqrt{6}$	$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$
n prime?	yes	yes	yes	no ($= 8\ell$)
$3 \mid (n - 1)$	yes	yes	yes	yes (for ℓ)
p cube mod n ?	yes	yes	no	yes (mod ℓ)
Howe (H1)	✓	✓	✓	✓
Howe (H2)	✓	✓	✓	mixed
Howe (H3)	1	1	1	4
(2,2) cover exists?	yes	yes	yes	via ℓ -restriction
Cover-based attack	no (B5)	no (B5)	no (B5)	no (B5)
Verdict	PASS	PASS	PASS	PASS

Table 1: Cross-curve structural-audit results.

Surprising auxiliary finding. Among the four curves, three (secp256k1, P-256, Curve25519) satisfy $p^{(n-1)/3} \equiv 1 \pmod{n}$ — i.e., p is a cube in $(\mathbb{Z}/n)^*$, giving embedding degree $(n - 1)/3$ rather than $n - 1$. brainpoolP256r1 is the outlier. This is not security-relevant (embedding degree remains ≥ 253 bits in every case) but is a notable structural coincidence among the deployed curves.

Curve25519’s cofactor changes Howe but not the verdict. Curve25519 has cofactor 8, so $\gcd(n, n_{\text{twist}}) \geq 4 > 1$ and Howe (H3) fails on the full $\#E$. Restricting to the prime subgroup of size $\ell = 2^{252} + 27742 \dots$, the framework applies and the verdict is the same.

8 The structural-completeness theorem

Theorem 8.1 (Structural Completeness for Prime-Field Ordinary ECC). *Let E be an ordinary elliptic curve over a prime field $\mathbb{F}_p$, with prime-order subgroup of size r (i.e., $\#E(\mathbb{F}_p) = h \cdot r$ with $h \in \{1, 2, 4, 8\}$ and r prime). Let $\mathcal{A}$ be any algorithm in the attack family*

$$\mathcal{A} \in \{\mathbb{F}_p\text{-isogeny transport, Cl(End)-orbit walking, } (N, N)\text{-cover, higher-genus cover, supersingular-reduction, VFCG-}\rho\}$$

producing a candidate value $\tilde{d}$ for the secret d . Then

$$T(\mathcal{A}) := \text{expected time of } \mathcal{A} \text{ to produce correct } \tilde{d} \geq \sqrt{r/|\text{Aut}(E)|}$$

with $|\text{Aut}(E)| \in \{2, 4, 6\}$ the size of the curve's automorphism group.

Proof. Case analysis over $\mathcal{A}$:

- **$\mathbb{F}_p$ -isogeny transport:** by B1 (Tate), the transported instance has the same r ; the algorithm reduces to plain ρ on the transported curve, also $\sqrt{r}$ cost.
- **Cl(End)-orbit walking:** by B2 ($h = 1$ at the crater for small-CM curves) or by orbit-walk-cost $\geq$ saving for $h > 1$, the algorithm reduces to plain ρ .
- **(N, N) -cover via Howe-gluing:** by B4–B5, the cover's Jacobian has genus 2 and DLP cost $O(p) > O(\sqrt{r})$. For secp256k1 specifically, Corollary 6.4 sharpens this to $O(p^{3/2}) \approx 2^{384}$ because Proposition 6.3 forces the cover's field of definition to be $\mathbb{F}_{p^3}$. Algorithm cost dominated by $\text{Jac}(C)$ DLP.
- **Higher-genus cover via Weil restriction:** by B5–B6. Genus $g \geq 2$ covers have DLP cost $O(p^{2-2/g}) \geq O(p) > O(\sqrt{r})$ over prime fields; Diem 2011 sub-exp is structurally blocked.
- **Supersingular-reduction attack:** by B7. The supersingular isogeny problem is disjoint from $\mathbb{F}_p$ ECDLP; canonical-lift requires anomalous curves.
- **VFCG- ρ :** by Section 5. The c_γ -in-small-subgroup property is satisfied uniformly but does not yield a walk speedup.

In each case, $T(\mathcal{A}) \geq \sqrt{r}/\sqrt{|\text{Aut}(E)|}$, which is the plain Pollard ρ cost. $\square$

What the theorem says and does not say

Says:

- For every deployed ECC curve in the prime-field ordinary regime, the seven specific isogeny-graph attack families do not yield speedups over plain ρ .
- The result holds under standard assumptions: Tate's theorem, Howe's theorem, Gaudry's index-calculus complexity, Diem-Thomé's refinement, and the standing assumption that prime-field hyperelliptic DLP is not sub-exponentially solvable.

Does not say:

- That ECDLP is provably hard (no such result is known under any natural complexity assumption).
- That side-channel / HNP-style attacks (scalar-side) are inapplicable.
- That post-quantum quantum-Shor attacks fail.
- That no future cryptanalytic technique will fall outside the seven enumerated families.

9 Open questions

In-scope: refinements of the theorem

- Tightening B5: is the $O(p^{2-2/g})$ Gaudry bound tight? (The (ℓ, ℓ) -universality in Corollary 4.3 strengthens the existing argument but the asymptotic remains the same.)
- Beyond hyperelliptic: do non-hyperelliptic genus- g curves admit faster DLP algorithms over $\mathbb{F}_p$?
- The tightness of Lemma 4.1: the obstruction holds for all $p \not\equiv 0, \ell-1 \pmod{\ell}$ (Exp Z', 2026-07-06; see `exp_zprime2_generalized_lemma.py`). Counterexamples exist precisely when $p \equiv \ell-1 \pmod{\ell}$; in that case $\ell \mid \#E^t$ as well, but B5 (Corollary 4.3) still holds via Honda–Tate alone.
- Indirect uses of supersingular reductions for fault-attack- amplified ECDLP recovery on isogenous siblings: speculative.
- Exact Cardona–Quer normalisation of Igusa invariants for C_{43} (cross-verification with Sage / Magma pending).
- *Resolved (Proposition 4.6)*: For any genus-2 Jacobian with biquadratic Weil polynomial $T^4 + a_2T^2 + p^2$ and $p \nmid a_2$, the prime p always *splits* in the associated quadratic field $K = \mathbb{Q}(\sqrt{\text{sf}(a_2^2 - 4p^2)})$. This was left as an empirical pattern in earlier versions; it is now proved algebraically and verified for 82 cases.

Out-of-scope: separate attack programmes

- Side-channel / HNP — where every real-world ECDSA break of the past 15 years lives. Includes the proposed novel GLV-aware HNP variant.
- Quantum cryptanalysis (Shor, Grover).
- SQIsign / supersingular-isogeny-based PQC cryptanalysis.
- Implementation cryptanalysis (fault, parsing, BIGNUM bugs).

10 Reproducibility

All scripts are PARI/GP 2.17+ scripts with committed expected outputs. Total runtime approximately 3 minutes on an M-series Mac.

File	LOC
audit.gp	235
cl_o_horizontal.gp	203
structural_invariants.gp	292
vfcg_experiment.gp	254
vfcg_variants.gp	284
howe_gluing_test.gp	260
howe_explicit_cover.gp	274
cover_complexity.gp	165
supersingular_reductions.gp	171
p256_comparison.gp	283
multi_curve_audit.gp	274
curve_audit_template.gp	120
generate_cm_curve.gp	80
glv_hnp_phase1.gp	180
fp3_obstruction_secp256k1.gp	200
igusa_f43_howe.gp	58
Total	≈ 3533

In addition, a Rust integration test (`tests/curve_audit.rs`, ≈ 190 lines) runs the audit via `cargo test` on every `CurveParams` constant defined in the codebase, asserting structural completeness for each.

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